

# GAOSHENG LIU

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## RESEARCH INTERESTS

Embedded systems and software, low-power wireless sensor networks, Internet of Things (IoT), battery-free and energy-harvesting systems, intermittent computing, edge AI and TinyML, communication protocol design, cyber-physical systems, AIoT, and wearable computing.

## EDUCATION

**Ph.D. in Computer Science, Vrije Universiteit Amsterdam** 2021 – 2025
*Advisors: Prof. Henri Bal and Prof. Lin Wang*
Thesis: Communication and Data Aggregation in Battery-Free IoT Networks

**M.Eng. in Computer Science, Tianjin University** 2016 – 2019
Focus: Secure communication in wireless sensor networks

**B.Eng. in Software Engineering, Tianjin University of Technology and Education** 2011 – 2015

## PROFESSIONAL EXPERIENCE

**Researcher, Missouri University of Science and Technology ), USA** Oct 2026 – now *Advisor: Sajal k. Das*

**Visiting Researcher, University of Trento, Italy** Jun 2026 – Jun 2027
*Advisor: Prof. Kasim Sinan Yildirim*

**Postdoctoral Researcher, University of Trento, Italy** Jun 2025 – Jun 2026
*Advisor: Prof. Kasim Sinan Yildirim*
Research on communication protocols and TinyML inference for battery-free IoT systems. Funded by NextGenerationEU project (E63C22002070006, €34,000). Designed MSP430 and STM32WL33CC1 firmware.

**Researcher, Vrije Universiteit Amsterdam, Netherlands** Jan 2025 – May 2025
Continued post-PhD research on battery-free networking protocols and system evaluation.

**Software Engineer, Huawei Technologies, Beijing, China** Apr 2019 – Jan 2021
IPS/Vehicle network software development. Full-cycle engineering: requirements, design, implementation, testing, and deployment for networking equipment.

## PUBLICATIONS

*Under Submission*

G. Liu. **Rethinking Battery-free Sensing Communication via Wake-up Radios.** *Under submission, 2026.*

*Journal Papers*

G. Liu, K.S. Yildirim, and L. Wang. **FreeBeacon: Efficient Communication and Data Aggregation in Battery-Free IoT.** *IEEE TMC, vol. 25, no. 3, pp. 3182–3195, 2026.* **[CCF-A, JCR Q1, IF=9.2]**

G. Liu and L. Wang. **Data on the Go: Seamless Data Routing for Intermittently-Powered Battery-Free Sensing.** *IEEE TMC, vol. 23, no. 12, pp. 13406–13419, 2024.* **[CCF-A, JCR Q1, IF=9.2]**

*Conference Papers*

M. Gottardelli, G. Liu, K.S. Yildirim. **Architecture Adaptation for Pretrained Models on Constrained Edge AI Accelerators.** *EdgeSys 2026.*

G. Liu, V. Nigade, H. Bal, and L. Wang. **A Little Certainty Is All We Need: Discovery and Synchronization Acceleration in Battery-Free IoT.** *ACM APNet 2024, Sydney, pp. 38–44.*

G. Liu and L. Wang. **Routing for Intermittently-Powered Sensing Systems.** *IEEE IPCCC 2023, pp. 274–282.* **[Best Paper Award, 1/69]**

G. Liu and L. Wang. **Self-Sustainable Cyber-Physical Systems with Collaborative Intermittent Computing.** *ACM e-Energy 2021, pp. 316–321.*

G. Liu, X. Wang, X. Li, J. Hao, and Z. Feng. **ESRQ: An Efficient Secure Routing Method in Wireless Sensor Networks Based on Q-Learning.** *IEEE TrustCom 2018, pp. 149–155.*

## HONOURS AND AWARDS

**IPCCC 2023 Best Paper Award (sole award, 1/69)** 2023

**ACM SenSys 2024 Student Travel Grant ($1,000)** 2024

**ACM SIGCOMM 2026 Student Travel Grant ($1,000)** 2026

## ACADEMIC SERVICE

Reviewer: IEEE Transactions on Mobile Computing (2025), IEEE Network guest editor (2026), IEEE Transactions on Sustainable Computing (2026), MobiQuitous 2026 TPC, IEEE MASS 2026 reviewer, IMC 2026 shadow reviewer, INFOCOM 2027

Artifact Evaluation Committee: ACM ATC'26, USENIX NSDI 2026 (Spring), ACM MobiSys 2025, IEEE/ACM MICRO 2025, ACM MobiSys 2023

## TECHNICAL SKILLS

**Embedded Hardware & Platforms**

- MCUs: TI MSP430FR5994/FR6989, STM32WL33CC1, Nordic nRF52840, Riotee
- PCB design: KiCad 9.0 (schematic capture, 4-layer PCB layout, BOM management)
- Components: Energy harvester, Sensors, FRAM
- Interfaces: SPI, I2C, UART, ADC/DAC, GPIO; oscilloscope and logic analyzer debugging

**Software & Programming**

- Embedded C: bare-metal and HAL-based firmware development (FreeRTOS, TI DriverLib, STM32 HAL)
- Python: simulation frameworks, data analysis, ML prototyping (PyTorch, NumPy, Matplotlib)
- Other: Java, MATLAB, OMNeT++ (discrete event simulation)
- Tools: Git, Docker, LaTeX, Linux, Make/CMake

**Wireless & Networking**

- Protocols: BLE 5.x, IEEE 802.15.4, Sub-GHz (MRSUBG), wake-up radio (WuR)
- Networking: protocol design for intermittent/energy-harvesting networks, multi-hop routing, data aggregation
- Industry: IPS/Vehicle core network software (Huawei, 2 years)

**AI & Edge Intelligence**

- TinyML / MCUNet: neural network deployment on microcontrollers
- Architecture search: hardware-aware model adaptation for constrained edge accelerators
- Frameworks: PyTorch, TensorFlow Lite Micro

## REFERENCES

**Prof. Henri Bal,** Vrije Universiteit Amsterdam | h.e.bal@vu.nl *(PhD supervisor)*

**Prof. Lin Wang,** University of Paderborn | lin.wang@uni-paderborn.de *(PhD co-supervisor)*

**Prof. Kasim Sinan Yildirim,** University of Trento | kasimsinan.yildirim@unitn.it *(Postdoc supervisor)*

**Prof. Sajal Das,** Missouri University of Science and Technology | sdas@mst.edu *(Postdoc supervisor)*